Calculus Calculator Program — Complete Guide

TI-Nspire Toolkit · IvyTutoring.net · Practice Questions & Explanations for Every Function

How to use: Select a tool from the menu, enter values, press Enter. All tools use the TI-Nspire CAS engine for symbolic computation. Press Esc to go back. This guide covers AP Calculus AB and BC.

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1. Limits

Evaluate Limit

Computes lim f(x) as x→a using CAS. Works for indeterminate forms (0/0, ∞/∞) via L'Hôpital.

Inputs: `f(x)= sin(x)/x` `x → 0`

PRACTICE QUESTION

Find lim $(x^2 - 4)/(x - 2)$ as x → 2.

Enter: `f(x)= (x^2−4)/(x−2) x → 2`

Answer: 4

One-Sided Limits

Computes left-hand limit ($x \to a^-$), right-hand limit ($x \to a^+$), and two-sided limit.

Inputs: `f(x)= 1/x` `x → 0`

PRACTICE QUESTION

Evaluate: lim |x|/x as x → 0 from the left and right.

Enter: `f(x)= abs(x)/x x → 0`

Answer: Left = –1, Right = 1, Two-sided DNE

Limit at Infinity

Finds lim f(x) as x→+∞ and x→−∞. Also identifies horizontal asymptotes.

Input: `f(x)= (3x^2+1)/(x^2−5)`

PRACTICE QUESTION

What is the horizontal asymptote of f(x) = (5x − 1)/(2x + 3)?

Answer: y = 5/2 (lim as x → ±∞ is 5/2)

Continuity Test

Three-step check: f(a) exists, lim exists, f(a) = lim.

Inputs: `f(x)= x^2` `at x= 3`

PRACTICE QUESTION

Is f(x) = (x² − 1)/(x − 1) continuous at x = 1?

Enter: `f(x)= (x^2−1)/(x−1) at x= 1`

Answer: f(1) undefined, lim = 2 → NOT continuous

📖 Limit Laws

Sum, product, quotient rules · Squeeze Theorem · L'Hôpital's Rule · Common limits: sin(x)/x → 1, $(1+1/n)^n$ → e

2. Derivatives

Derivative

Symbolic first and second derivatives via CAS. Shows simplified form.

Input: `f(x)= x^3+2x`

PRACTICE QUESTION

Find $f'(x)$ for $f(x) = 3x^4 - 2x^2 + 5x$.

Answer: $f'(x) = 12x^3 - 4x + 5$

n-th Derivative

Computes derivatives up to the n-th order (max 6 shown).

Inputs: `f(x)= sin(x)` `n= 4`

PRACTICE QUESTION

What is the 4th derivative of sin(x)?

Answer: $f^{(4)}(x) = \sin(x)$ (cycles every 4)

Derivative at a Point

Evaluates f(a) and f′(a) — the slope at a specific x-value.

Inputs: `f(x)= x^2+1` `at x= 3`

PRACTICE QUESTION

What is the instantaneous rate of change of $f(x) = \sqrt{x}$ at $x = 9$?

Enter: `f(x)= sqrt(x) at x= 9`

Answer: $f'(9) = 1/(2\sqrt{9}) = 1/6$

Implicit Differentiation

Finds dy/dx from an implicit equation using impDif().

Input: `x^2+y^2=25`

PRACTICE QUESTION

Find dy/dx for $x^3 + y^3 = 6xy$.

Enter: x^3+y^3=6x∗y

Answer: $dy/dx = (6y - 3x^2)/(3y^2 - 6x)$

📖 Derivative Rules

Power, Product, Quotient, Chain rules · $d/dx[e^x] = e^x$ · $d/dx[\ln x] = 1/x$ · $d/dx[a^x] = a^x \ln(a)$

📖 Trig Derivatives

$d/dx[\sin x] = \cos x$ · $d/dx[\cos x] = -\sin x$ · $d/dx[\tan x] = \sec^2 x$ · Inverse trig: $d/dx[\arctan x] = 1/(1+x^2)$

3. Applications of Derivatives

Tangent / Normal Line

Finds the tangent line y = mx + b and normal line (perpendicular) at a point.

Inputs: `f(x)= x^2` `at x= 2`

PRACTICE QUESTION

Find the equation of the tangent line to $f(x) = x^3$ at x = 1.

Enter: `f(x)= x^3 at x= 1`

Answer: y = 3x – 2 (slope = 3, point (1, 1))

Critical Points & Extrema

Sets $f'(x) = 0$, finds zeros, applies 2nd Derivative Test to classify each as local max, min, or inconclusive.

Input: `f(x)= x^3−3x`

PRACTICE QUESTION

Find the local extrema of $f(x) = x^3 - 12x$.

Enter: x^3−12x

Answer: x = –2 → local max (f = 16), x = 2 → local min (f = –16)

Inflection Points

Sets $f''(x) = 0$, finds where concavity changes.

Input: `f(x)= x^4−6x^2`

PRACTICE QUESTION

Find the inflection points of $f(x) = x^3 - 3x^2 + 2$.

Enter: x^3−3x^2+2

Answer: Inflection at (1, 0); f″ = 6x – 6, f″ = 0 at x = 1

Mean Value Theorem

Finds the average rate of change on [a,b] and solves f′(c) = average rate to find c.

Inputs: `f(x)= x^2` `a= 1` `b= 3`

PRACTICE QUESTION

For f(x) = x² on [0, 4], find c where f′(c) equals the average rate.

Answer: Average = (16–0)/4 = 4, f′(c) = 2c = 4, so c = 2

Linear Approximation

L(x) = f(a) + f′(a)(x – a). Estimates f(x) near x = a.

Inputs: `f(x)= sqrt(x)` `at x= 4` `estimate x= 4.1`

PRACTICE QUESTION

Use linear approximation to estimate √4.1.

Enter: `f(x)= sqrt(x) at x= 4 estimate x= 4.1`

Answer: L(4.1) = 2 + (1/4)(0.1) = 2.025 (exact: 2.0248…)

Related Rates Helper

Sets up an implicit differentiation step for related rates problems.

Inputs: `Equation= x^2+y^2=25` `dx/dt= 3` `at x= 3`

PRACTICE QUESTION

A circle has x² + y² = 100. If dx/dt = 2 when x = 6, find dy/dt.

Answer: dy/dx = –x/y = –6/8, dy/dt = (–6/8)(2) = –1.5

📖 Optimization Steps

1. Diagram → 2. Identify max/min → 3. Objective function → 4. Reduce to 1 variable → 5. f′ = 0 → 6. Verify → 7. Answer

4. Integrals

Indefinite Integral

Symbolic antiderivative via CAS. Returns F(x) + C.

Input: `f(x)= x^2+3x`

PRACTICE QUESTION

Find $\int(6x^2 - 4x + 1)\,dx$.

Enter: 6x^2−4x+1

Answer: $2x^3 - 2x^2 + x + C$

Definite Integral

Exact symbolic result + decimal approximation. Shows antiderivative.

Inputs: `f(x)= x^2` `a= 0` `b= 3`

PRACTICE QUESTION

Evaluate $\int_0^2 (x^3 - x)\,dx$.

Enter: x^3−x a=0 b=2

Answer: $[x^4/4 - x^2/2]_0^2 = 4 - 2 = 2$

u-Substitution

Shows the substitution steps: u, du/dx, du. Then integrates via CAS.

Inputs: `f(x)= 2x*cos(x^2)` `u= x^2`

PRACTICE QUESTION

Find $\int 3x^2 \cdot e^{x^3}\,dx$ using $u = x^3$.

Enter: 3x^2*e^(x^3) u= x^3

Answer: $du = 3x^2dx \rightarrow \int e^u\,du = e^{x^3} + C$

Integration by Parts

∫u dv = uv − ∫v du. Shows u, du, dv, v breakdown and final result.

Inputs: `f(x)= x*e^x` `u= x` `dv= e^x`

PRACTICE QUESTION

Find ∫ x·sin(x) dx.

Enter: `x*sin(x)` `u= x` `dv= sin(x)`

Answer: −x·cos(x) + sin(x) + C

📖 Integration Rules & Trig Integrals

Power rule, $\int e^x = e^x$, $\int 1/x = \ln|x|$, trig integrals, LIATE mnemonic for by-parts, trig substitution patterns

5. Applications of Integrals

Area Between Curves

Area = $\int |f(x) - g(x)|\, dx$ from a to b.

Inputs: `f(x)= x^2` `g(x)= x` `a= 0` `b= 1`

PRACTICE QUESTION

Find the area between $y = x^2$ and $y = x$ from $x = 0$ to $x = 1$.

Answer: $\int_0^1 (x - x^2)\, dx = [x^2/2 - x^3/3]_0^1 = 1/6$

Volume: Disk/Washer Method

$V = \pi\int[R(x)^2 - r(x)^2]\, dx$. Enter outer and inner radii.

Inputs: `R(x)= sqrt(x)` `r(x)= 0` `a= 0` `b= 4`

PRACTICE QUESTION

Find the volume when $y = \sqrt{x}$ is rotated about the x-axis from 0 to 4.

Answer: $V = \pi\int_0^4 x\, dx = \pi[x^2/2]_0^4 = 8\pi$

Volume: Shell Method

$V = 2\pi\int x \cdot f(x)\, dx$. For rotation about the y-axis.

Inputs: `f(x)= x^2` `a= 0` `b= 2`

PRACTICE QUESTION

Find the volume when $y = x^2$ $(0 \le x \le 2)$ is rotated about the y-axis.

Answer: $V = 2\pi\int_0^2 x^3\, dx = 2\pi[x^4/4]_0^2 = 8\pi$

Average Value of Function

$f_{avg} = 1/(b-a) \cdot \int f(x)\, dx$ from a to b.

Inputs: `f(x)= x^2` `a= 0` `b= 3`

PRACTICE QUESTION

Find the average value of $f(x) = \sin(x)$ on $[0, \pi]$.

Enter: `sin(x) a= 0 b= pi`

Answer: $(1/\pi)\cdot\int_0^p \sin(x)\, dx = (1/\pi)(2) = 2/\pi \approx 0.637$

Arc Length

$L = \int \sqrt{1 + [f'(x)]^2}\, dx$ from a to b.

Inputs: `f(x)= x^(3/2)` `a= 0` `b= 4`

PRACTICE QUESTION

Find the arc length of y = x^(3/2) from x = 0 to x = 4.

Answer: $f' = (3/2)x^{(1/2)}$, $L = \int_0^4 \sqrt{1 + 9x/4}\, dx \approx 9.073$

📖 Fundamental Theorem of Calculus

FTC 1: $d/dx \int_a^x f(t)dt = f(x)$ · FTC 2: $\int_a^b f(x)dx = F(b) - F(a)$ · MVT for Integrals: f(c) = f_avg

6. Series & Sequences (BC Topics)

Taylor Polynomial

Generates the n-th degree Taylor polynomial about x = a using CAS taylor().

Inputs: `f(x)= e^x` `about a= 0` `order= 5`

PRACTICE QUESTION

Find the 4th-degree Maclaurin polynomial for cos(x).

Enter: `cos(x) a= 0 order= 4`

Answer: $P_4(x) = 1 - x^2/2 + x^4/24$

Partial Sum of Series

Computes Σ a(n) from n = a to n = b using CAS sum().

Inputs: `a(n)= 1/n^2` `from n= 1` `to n= 100`

PRACTICE QUESTION

Estimate $\Sigma\ 1/n^2$ from n = 1 to 1000 (should approach $\pi^2/6 \approx 1.6449$).

Answer: $S \approx 1.6439$ (exact limit = $\pi^2/6$)

Geometric Series

Finite sum $S(n) = a(1-r^n)/(1-r)$. If $|r| < 1$, shows $S(\infty) = a/(1-r)$.

Inputs: `a= 1` `r= 0.5` `n= 10`

PRACTICE QUESTION

Find the sum of 3 + 3/2 + 3/4 + 3/8 + … (infinite geometric series).

Enter: `a= 3 r= 0.5`

Answer: $S(\infty) = 3/(1-0.5) = 6$

Power Series

Expands f(x) as a power series (6 terms) using CAS taylor().

Inputs: `f(x)= 1/(1-x)` `about a= 0`

PRACTICE QUESTION

Express e^x as a power series about x = 0 (6 terms).

Enter: e^x a= `0`

Answer: $1 + x + x^2/2 + x^3/6 + x^4/24 + x^5/120 + \ldots$

📖 Convergence Tests

n-th Term · Geometric (|r|<1) · p-Series (p>1) · Integral · Comparison · Limit Comparison · Ratio · Alternating Series

7. Differential Equations

Solve ODE

Uses CAS deSolve() to find the general or particular solution of a first-order ODE.

Inputs: `ODE= y'=2x` `IC: y(0)=1`

PRACTICE QUESTION

Solve dy/dx = $3x^2$ with y(0) = 5.

Enter: `y'=3x^2 IC: y(0)=5`

Answer: y = x^3 + 5

Euler's Method

Step-by-step numerical ODE approximation. Shows (x, y) and slope at each step.

Inputs: `dy/dx= x+y` `x0= 0` `y0= 1` `h= 0.1` `steps= 5`

PRACTICE QUESTION

Use Euler's method with h = 0.5 and 4 steps for dy/dx = y, y(0) = 1.

Enter: `dy/dx= y x0=0 y0=1 h=0.5 steps=4`

Answer: y(2) ≈ 5.0625 (exact: e^2 ≈ 7.389)

Separation of Variables

Solves separable ODEs using CAS deSolve().

Input: `dy/dx= y*x`

PRACTICE QUESTION

Solve dy/dx = y·cos(x).

Enter: `y*cos(x)`

Answer: y = Ce^(sin(x))

📖 Common Differential Equations

Separable · Linear 1st order with integrating factor · Exponential growth/decay dy/dt = ky · Logistic dy/dt = ky(1−y/L) · Slope fields

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